Agricultural Systems 171 (2019) 13–22

Contents lists available at ScienceDirect

Agricultural Systems

journal homepage: www.elsevier.com/locate/agsy

Soil carbon saturation, productivity, and carbon and nitrogen cycling in crop-pasture rotations

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ARTICLE INFO

Keywords:
Soil organic matter
Agroecosystem modeling
Long-term experiments
Crop-pasture interseeding

ABSTRACT

Agricultural systems integrating perennial grass-legume pastures in rotation with grain crops sustain high crop yields while preserving soil organic carbon (C_s) with low nitrogen (N) fertilizer inputs. We hypothesize that C_s saturation in the topsoil may explain the favorable C and N cycling in these systems. We tested this hypothesis by evaluating and simulating three contrasting crop and pasture rotational systems from a 20-year no-till experiment in Treinta y Tres, Uruguay. The systems were: 1) Continuous annual cropping (CC); 2) crop-pasture rotation with two years of crops and four years of pastures (CP); and 3) perennial pasture (PP). Using the Cycles agroecosystems model, we evaluated the inclusion or exclusion of a C_s saturation algorithm. The model simulated forage, soybean, and sorghum grain yields correctly, with low root mean square error (RMSE) of 1.5, 0.7 and 1.0 Mg ha^{-1}, respectively. Measurements show C_s accretion and C_s decline for the first and second half of the experiment, respectively. The C_s accretion rate was highest for PP, while the C_s decline was highest for CC (1.3 vs −0.6 Mg ha^{-1} y^{-1} of C). The model captured this C_s dynamics and performed better when using the C_s saturation algorithm than when excluding it (RMSE 4.7 vs 6.8 Mg C ha^{-1} and relative RMSE of 14% and 21% for the top 15-cm). The model with saturation simulated subsoil C_s distribution with depth well, and simulated faster N turnover and greater N availability for the subsequent grain crop in CP vs CC. The results suggest that C_s saturation, and by extension soil organic N saturation, underpin the sustainability of crop-pasture rotations, and that modeling C_s saturation dynamics can be critical to reliably simulate complex crop-pasture rotational systems.

1. Introduction

Sustainable intensification of grain and livestock production requires balancing high productivity with soil, water, and biodiversity preservation (Matson et al., 1997; Cassman, 1999). Integrated crop and livestock production systems that include a perennial grass-legume multi-year phase in the rotation can increase landscape diversity and match or exceed the performance of less diverse systems (Davis et al., 2012). These systems are common in Australia (Hochman et al., 2013), and subtropical and temperate South-America (Carvalho et al., 2010; Moraes et al., 2014; Franzluebbers et al., 2014). Understanding the mechanisms by which integrated crop-pasture systems operate can provide pathways for the sustainable intensification of agriculture in other regions.

Integrated crop-livestock systems diversify production and risk, reduce costs, and can stabilize farm economic returns (García-Préchac et al., 2004; Carvalho et al., 2010; Moraes et al., 2014), although the latter depends on the relative price of grain and animal products (Ernst et al., 2016). In these systems, the plant species and the length of annual cropping and pasture phases are selected and managed to express beneficial ecological synergies, where nitrogen (N) requirements of non-legume crops are partially fulfilled by biological N fixation (BNF) during the perennial pasture phase (Carvalho et al., 2010; Franzluebbers et al., 2014; Soussana and Lemaire, 2014). Livestock in the system provides environmental benefits through the return of animal manure, enhancing soil tilth, fertility, and carbon (C) sequestration (Russelle et al., 2007). At the landscape level, and due to the representation of different phases of the rotation in different fields, farms and watersheds become a rich mosaic of annual and perennial crops. These landscapes contrast with the simplified and input-intensive agroecosystems that dominate areas like the Midwestern United States. Such simplified systems can produce many

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https://doi.org/10.1016/j.agsy.2018.11.001
Received 30 December 2017; Received in revised form 2 November 2018; Accepted 2 November 2018
Available online 12 January 2019
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undesired outcomes, such as high vulnerability to pest and weed invasion, water pollution, and greenhouse gas emission (Matson et al., 1997; Robertson and Swinton, 2005; Alexander et al., 2007).

One of the signatures of crop-pasture rotational systems is the oscillation of the soil organic C (C_s) and soil organic N (N_s) throughout the rotation: C_s and N_s decrease during the annual cropping phase and recover quickly during the perennial pasture phase (Díaz Rosello, 1992a; Díaz Rosello, 1992b; Díaz-Zorita et al., 2002). The C_s and N_s recovery during the pasture phase is explained by the large inputs of C and N from perennial plant species in the pasture phase: perennial species contribute three to seven times more C and N to the litter pool than annual species through greater root production (Mapfumo et al., 2002; Dupont et al., 2014); these roots are also placed deeper in the soil profile (Monti and Zatta, 2009). Most of the N_s gain is due to BNF. During the annual phase, N_s accumulated during the pasture phase is gradually mineralized, reducing non-legume annual crop N fertilizer needs. This gradual N mineralization is believed to improve the synchrony between N supply and N uptake, reducing opportunities for N losses and enabling higher productivity. When pastures are removed from the rotation, changes in N supply and other soil factors may result in a decreasing yield ceiling and a higher yield gap even when annual crops are managed under no-till, as documented for wheat (Ernst et al., 2016).

We propose that this improved N supply can be explained by a mechanism rooted in the C_s saturation theory. This theory states that mineral soils have a limited capacity for C_s stabilization, determined by the concentration of silt and clay (Hassink and Whitmore, 1997). Under this theory, the rate of C_s storage tends to zero when, driven by increasing C inputs or reducing C_s mineralization, C_s approaches saturation (Six et al., 2002; Kemanian et al., 2005; Stewart et al., 2008). Because of the relatively narrow range of the soil organic matter C:N ratio, C_s saturation can lead to N_s saturation (Aber et al., 1989; Lovett and Goodale, 2011). Under saturation conditions, new organic residue additions and relatively labile soil organic matter would not be protected from microbial decomposition by soil aggregates or nano-scale organo-mineral associations, all of which results in a higher overall microbial turnover rate. The higher turnover rate also causes a higher N_s mineralization rate (Castellano et al., 2012; White et al., 2014). In addition, this process can trigger a positive feedback between N supply and plant productivity (Baer and Blair, 2008): high productivity accelerates progress towards C_s saturation, and C_s saturation ensures higher productivity due to a consistent supply of N.

Soils in the pasture phase of a crop-pasture rotation have some of the attributes of C_s saturated soils, and it is reasonable to assume that several desirable attributes of these systems derive from soil C and N cycling in a soil matrix with, at least, partial C_s and N_s saturation. While traditionally, the benefits of pastures in agricultural systems have been ascribed to diversification through livestock production, erosion control, and BNF, maintaining some soil layers near C_s saturation might be one of the critical services provided by perennials in the system. Despite its potential importance, modeling C_s and N_s saturation in systems that include pastures has not been previously attempted.

Earlier usage of the saturation concept in models suggests that the saturation ratio matters, not only near the saturation boundary, but along the entire saturation range (Kemanian et al., 2005; Kemanian and Stöckle, 2010; Heitkamp et al., 2012; Nicoloso et al., 2016). For example, White et al. (2014) have shown that the higher the saturation ratio the higher the release of N from mineralization, even altering the critical C:N ratio of fresh residues that triggers mineralization. These modeling efforts explored systems that do not include the complexity of crop-pasture rotations, mostly because they are bounded by empirical data from continuous cropping experiments (Kemanian et al., 2011; Nicoloso et al., 2016), with only exceptional attempts to manipulate C_s saturation in controlled conditions (Stewart et al., 2008). Work at the microbial and soil aggregate scale (Ahrens et al., 2015) is difficult to translate to the agroecosystem scale. Understanding both the agroecosystem-scale mechanisms by which C_s and N_s saturation operate, and how these contribute to the long-term sustainability of crop-pasture rotations, can be fundamental to chart sustainable intensification pathways.

The goal of this work is to test if including C_s saturation and by extension N_s saturation in an agroecosystems simulation model allows for more accurate modeling of the observed dynamics of C and N and plant productivity in crop-pasture rotational systems. We report C_s dynamics and crop and pasture performance from a 20-year experiment in Treinta y Tres, Uruguay with three rotational systems, and compare these results with those obtained with the simulation model Cycles. The rotational systems are an annual continuous cropping system, a crop-pasture rotational system, and a perennial pasture that has not been tilled since the beginning of the experiment. We tested if including or excluding a C_s saturation mechanism in the model Cycles alters the representation of both the C_s and N_s long-term dynamics and crop productivity. Our hypotheses are that including C_s saturation will enable more accurate simulation of: 1) the long-term C_s dynamics; 2) the distribution of C_s with depth, and 3) the supply of N that determines productivity in crop-pasture rotations.

2. Methods

2.1. Long-term experiment

In 1995, the National Institute of Agricultural Research of Uruguay (INIA) established a long-term crop-pasture rotational experiment at the Palo a Pique Experimental Unit (UEPP; 33°15′44 S, 54°29′ W). The UEPP is located in the subtropical humid region of the South American Pampas. The climate is mesothermic humid with a mean daily temperature of 23 and 11 °C in summer and winter, respectively, and a mean annual rainfall of 1300 mm distributed on average uniformly throughout the year, but with large inter-annual variations (Table 1).

Table 1
Weather monthly summary for Palo a Pique experimental site between 1995 and 2014.

Month	Solar	Precipitation	Penman-Monteith	Temperature			Wind speed
	Radiation		Reference ET	Maximum	Minimum	Dew Point	
	MJ m^{-2} d^{-1}	mm $month^{-1}$	mm $month^{-1}$	°C			m s^{-1}
January	24.8	91	182	30	17	14	2.5
February	20.7	131	137	28	17	14	2.3
March	17.7	104	124	27	15	13	2.1
April	13.3	128	85	23	12	9	1.9
May	9.9	133	60	20	8	5	1.7
June	8.2	116	50	17	6	3	1.9
July	9.0	96	58	16	6	3	2.2
August	11.7	103	75	18	7	4	2.3
September	15.6	105	96	19	8	5	2.7
October	19.3	100	128	23	11	8	2.8
November	23.7	86	157	26	13	10	2.7
December	25.3	106	180	28	15	12	2.6

The research site has a 3% average slope. The loam soils are Oxyaquic Argiudolls according to USDA Soil Taxonomy, and have moderate fertility with a well-developed Bt horizon and no rocks.

The site had a short history of soybean cropping with conventional tillage in the 1980s followed by a pasture of *Lolium multiflorum* L., *Lotus corniculatus* L. and *Trifolium repens* L. This was invaded after a few years by native and exotic species, mainly *Cynodon dactylon* (L.) Pers. By the time the rotational experiment was started in 1995, the vegetation had transitioned back to the native grassland of mixed C_3 and C_4 species. We used three no-till rotational systems: 1) CC: continuous cropping rotation of two years; 2) CP: six-year crop-pasture rotation that alternates four years of grass-legume pasture with two years of annual crops identical to CC; and 3) PP: perennial pasture of introduced perennial legumes and grasses that was renewed every six years (killed with glyphosate and no-till re-planted). Until 2005 the experiment only included forage species; grain crops were introduced thereafter as explained below. The species planted in each system for each period are listed in Table 2. The experimental design is a case of the "cyclic rotation experiment" as described by Patterson (1964), with no synchronic replications but with the same crops in the sequences of the different rotations. The experiment occupies eight different fields: six fields represent all entry points of the crop-pasture system in every year, one field represents the continuous cropping rotation, and one field the perennial pasture. The average field size was originally 6 ha (200 m wide x 300 m long downslope), allowing for cattle grazing.

From 1995 to 2004, the systems were mostly cattle grazed or cut for forage. In 2005, all fields were split in two. In one of the 3-ha halves of each field (100 m wide x 300 m long downslope), annual grain crops substituted for annual forage crops. The data and simulations presented here address fields that have had grain production since 2005. Previous reports on this experiment can be found in Terra and García-Préchac (2001), García-Préchac et al. (2004), and Terra et al. (2006). A record of the management practices used in each field was used to specify model inputs.

2.2. Weather database

Maximum and minimum daily temperature, dew point temperature, wind speed and downward solar radiation were taken from the closest weather station at Paso de la Laguna Experimental Unit, 30 km away from UEPP experimental site. Rainfall was recorded at the experimental site. Records from an automatic weather station located at UEPP during 2012–2014 indicated that the daily weather at these two locations was very similar. Solar radiation was available at Paso de la Laguna, although there were gaps in the data series. Existing records agreed with those available at NASA-POWER; the latter database was used throughout the years for consistency.

2.3. Soil description

A complete description of the Oxyaquic Vertic Argiudolls at the UEPP site was available from soil survey records (Durán et al., 2006), sampled as series Alferez at the 1996 field survey for Uruguay as part of the US NCSS Soil Characterization Database (National Cooperative Soil Survey, 2018). This soil profile named ´Uruguay Site 13′ (pedon ID 96FN930013), located at the experimental site, was used to build the model soil input file (Table 3). Soil volumetric water content at saturation, field capacity and permanent wilting point were estimated using the pedotransfer functions reported by Saxton and Rawls (2006). Textural homogeneity in the top 18-cm reflects the history of tillage. Soil sampling of the top 15-cm of each field at the beginning of the experiment in 1995 provided C_s information to initialize the eight simulated fields. Deeper soil layers were initialized with the same soil organic matter for all systems (Table 3). We also used the same bulk density for all treatments in the top 15 cm, despite differences in organic matter. That may overestimate the initial C_s mass in the top layer of the fields with the highest C_s concentration, and underestimate the initial C_s mass in the top layer of the fields with the lowest C_s concentration.

Table 2
Cropping sequences at the rotation experiment. In the first decade (1995–2004), crops were produced for forage agriculture only, while the last decade (2005–2014) included both forage and grain cropping in continuous cropping and crop-pasture systems.

Period: 1995-2004. Forage cropping rotation components

	Rotation year and season											
	Year 1		Year 2		Year 3		Year 4		Year 5		Year 6	
System	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer
Continuous Cropping	WF	S	WF	SF	WF	S	WF	SF	WF	S	WF	SF
Crop-Pasture	WF	S	WF	SF	W + P	Perennial pasture: OG/TF + BTR + WCl						
Perennial Pasture	Mixed grass-legume pasture: RG + BTR + WCl											

Period: 2005-2014. Grain and forage cropping rotation components

	Year 1		Year 2		Year 3		Year 4		Year 5		Year 6	
System	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer
Continuous Cropping	WC	GS	WC	SB	WC	GS	WC	SB	WC	GS	WC	SB
Crop-Pasture	WC	GS	WC	SB	W + P	Perennial pasture: OG/TF + BTR + WCl						
Perennial Pasture	Mixed grass-legume pasture: TF + BTR + WCl											

References: WF: Winter forage crop for grazing: Annual ryegrass (*Lolium multiflorum* Lam.), oat (*Avena sativa*) or wheat (*Triticum aestivum* L), S: sorghum (*Sorghum bicolor* L. Moench) for grazing, SF: Summer forage crops for grazing or hay: Sudangrass (*Sorghum x drummondii* (Steud.) Millsp. & Chase) or foxtail millet (*Setaria italica* (L.) P. Beauv.), WC: Winter cover of ryegrass, avena or grain wheat, OG: Orchardgrass (*Dactylis glomerata* L.), TF: Tall fescue (*Lolium arundinaceum* (Schreb.) S.J. Darbyshire, or *Festuca arundincaea* Schreb.), BTR: Birdsfoot trefoil (*Lotus corniculatus* L.), WCl: White clover (*Trifolium repens* L.), RG: Annual ryegrass, GS: Grain Sorghum, SB: Soybean (*Glycine* max). W + P: Grain wheat interseeded with perennial pasture. Plus sign indicates interseeded species. Dash sign indicates alternative planting.

Table 3
Soil properties at the beginning of the experiment.

System	All	All	All	CC	CP	PP	CC	CP	PP
Depth	Clay	Sand	Bulk density	Soil organic matter			Soil carbon saturation ratio		
cm	%	%	$Mg\ m^{-3}$	$g\ kg^{-1}$			$kg\ kg^{-1}$		
0 to 4	20.7	39.2	1.16	33.4	43.2	33.6	0.67	0.87	0.68
4 to 9	20.7	39.2	1.16	23.4	30.1	25.5	0.47	0.61	0.51
9 to 18	19.6	39	1.39	20.8	26.8	22.7	0.42	0.55	0.46
18 to 41	46.9	24.4	1.39	15.4	15.4	15.4	0.23	0.23	0.23
41 to 61	48.3	22.2	1.34	7.4	7.4	7.4	0.11	0.11	0.11
61 to 71	45.8	22.8	1.37	5.2	5.2	5.2	0.08	0.08	0.08
71 to 99	40.4	28.5	1.42	2.6	2.6	2.6	0.04	0.04	0.04
99 to 125	44.4	37.2	1.73	2.1	2.1	2.1	0.03	0.03	0.03
125 to 141	47.3	39.3	1.55	1.7	1.7	1.7	0.03	0.03	0.03
141 to 172	32.6	48.6	1.55	0.7	0.7	0.7	0.01	0.01	0.01
172 to 190	28.9	55.7	1.57	0.3	0.3	0.3	0.01	0.01	0.01

CC: continuous cropping rotation of two years; CP: six-year crop-pasture rotation that alternates four years of grass-legume pasture with two years of annual crops; PP: perennial pasture of introduced perennial legumes and grasses.

2.4. Agroecosystems model

All systems were simulated using the Cycles agroecosystems model. Cycles is a process-based, multi-year, multi-crop, and multi-soil layer simulation model that runs at a daily time step, with hydrology simulated with an adaptive sub-daily time step. Several modules of Cycles have been incorporated from CropSyst (Stöckle et al., 2003) and C-Farm (Kemanian and Stöckle, 2010). The heat and water transport algorithms are adapted from Campbell (1985). Cycles has modules to represent plant growth based on radiation and transpiration use efficiency (Stewart et al., 2008), coupled C and N cycling (Kemanian et al., 2005; Kemanian and Stöckle, 2010; White et al., 2014), soil water infiltration and redistribution, and the effects of management practices on biogeochemical processes. The model version we used does not simulate soil erosion. Cycles can simulate monoculture rotations, polycultures and relay crops. The inputs required to run Cycles are: 1) latitude, elevation, and daily weather data; 2) layer-by-layer initial soil profile properties (layer thickness, texture, bulk density, hydraulic properties, organic matter); 3) crop sequence; and 4) management operations (fertilization, irrigation, residue addition, tillage, harvest).

2.4.1. Soil organic carbon cycling

The C_s dynamics module in Cycles considers the controls of C_s saturation capacity (Hassink and Whitmore, 1997) in C stabilization. When microbes decay, a fraction of the microbial C loss is added to the C_s pool. This fraction is analogous to the humification coefficient in the Hénin and Dupuis (1945) representation of the C_s gain due to plant residue decomposition. The term humification acknowledges the C transfer from the residues pool to the soil organic matter pool (without implying that it is composed of "humic" substances). In Cycles, this coefficient depends on the saturation ratio, i.e. the C_s concentration in relation to a putative C_s saturation capacity, or C_x (kg C kg^{-1}soil), *sensu*Hassink and Whitmore (1997):

$$C_x = 0.021 + 0.038\, f_c \tag{1}$$

where f_c is the fractional clay concentration of that layer.

When the soil is near saturation, the fractional stabilization of microbial detritus into C_s or humification rate approaches zero (Fig. 1) using the following functional equation:

$$h_c = h_x \left[1 - \left(\frac{C_s}{C_x}\right)^n\right] \tag{2}$$

where h_x is the maximum humification rate as a function of clay and h_c is the operative humification rate. Further discussion of the implications of this structure can be found in Kemanian et al. (2011) and White et al. (2014). Unique to Cycles is the use of the saturation ratio to regulate the soil organic matter turnover rate, k_S (d^{-1}):

$$k_s = f_e f_t k_x f_s \tag{3}$$

$$f_s = 1 - 1/\left[1 + \left(\frac{C_s/C_x}{0.22}\right)^3\right] \tag{4}$$

where f_s is a factor that accelerates the turnover as the saturation ratio increases, f_e is the environmental factor (0–1), f_t is the tillage factor (> 1), k_x is the maximum undisturbed C decomposition rate (0.00015 d^{-1}).

2.4.2. Algorithm without carbon saturation

To test if saturation improves C_s dynamics simulation, we modified Cycles to represent the traditional non-saturating approach by setting $h_c = h_x$ and and $f_s = 1$:

$$\frac{\partial C_s}{\partial t} = h_x C_i - f_e f_t k_x C_s \tag{5}$$

The cycling of N is coupled to that of C as presented by White et al. (2014), which includes explicitly the microbial pool.

2.4.3. Plant competition module

A new module was introduced in Cycles to simulate multi-species plant communities. This module enables simulating grass-legume mixed pastures, and dual purpose and nurse crops. For instance, wheat underseeded with perennial pasture species can be grazed and later harvested for grain, while the perennial pasture remains. Each plant component has species-specific physiological parameters. The competition and acquisition of resources (i.e., water, nutrient, and solar radiation) among species as well as the community botanical composition are emergent properties of the interaction of species, environment, and management. Algorithms for light interception in the competition module are presented in Appendix A.

2.4.4. Pastures and crop production calibration and evaluation

Parameters for forage and grain species were calibrated from information compiled from several sources, and are provided for each of the 13 species in Appendix B.

Simulated forage yields were compared to field data from the experimental site reported by Terra et al. (2001), Rovira and Bermúdez (2003), and measured by Pravia in the years 2008–2011 (unpublished). Pasture forage yield was summarized on an annual basis for comparison

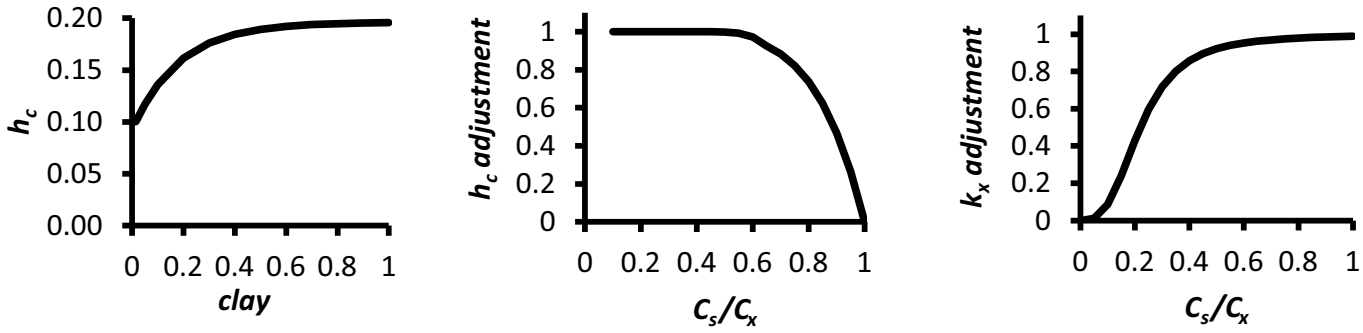


Fig. 1. Humification rate (h_c) and decomposition rate (k_x) relationship to soil clay content and soil carbon saturation ratio (C_s/C_x).

with the model outputs, for a total of 28 field-year combinations through the 20 years of simulations. For grain crops, 12 soybean and 13 sorghum field-years were available for model evaluation.

2.5. Soil organic carbon model evaluation

2.5.1. Soil organic carbon

Soil organic carbon model performance was evaluated using data from composites of 15-cm deep and 2.5-cm diameter soil samples taken from each field in each year. The sampling was stratified as described by Terra et al. (2006). Fields were divided in three pseudo-blocks 100-m long, each covering the upper slope, mid-slope, and bottom slope topographic positions, as determined by a detailed digital elevation map of the area (0.5 m vertical interval). Samples were taken every year in the fall (May), by randomly sampling and bulking 10 cores in georeferenced points of each pseudo-block. The split of the plots in 2005 was perpendicular to the pseudo-blocks, allowing the sampling by pseudo-block to continue.

Independent soil profile samples were taken in 2014 and 2015 using a soil core hydraulic cylinder probe of 4.7-cm diameter and 80-cm long. In 2014, 40 80-cm deep soil profiles were taken from the middle slope pseudo-block. After segmenting each profile by horizon (A1: 0–6 cm, A2: 6–18 cm, B: 18–40 cm) the samples were bulked within each field and horizon, and subsampled by horizon for analysis (A1, A2 and B), each analyzed by triplicate. In 2015, 60-cm deep soil profiles were taken from each pseudoblock with the same probe. Samples were separated in the field at 15-cm depth increments; the samples from the top 15-cm were used to calculate bulk density. For calculation of C_s stocks, we used the average bulk density of these samples for the top 15-cm of soil for both observed and simulated data (1.33 Mg m^{-3}). For deeper layers we used the bulk density reported in Table 3. Soil physical properties, particularly the hydrological properties, were kept constant during the simulations to isolate the putative effects of C and N cycling on crop performance.

2.5.2. Model statistical evaluation

Rotational system and year effects on observed C_s were tested with an analysis of variance. The year effect showed a significant quadratic term. We then calculated the root of the derivative of the quadratic model to find the year of the maximum C_s, which we used to split the period in two phases with linear responses. Model performance was evaluated using the root mean square error (RMSE), relative root mean square error (rRMSE) and coefficient of determination (r^2) between observed and simulated data. All analyses used the R software (R Core Team, 2013).

3. Results

3.1. Forage and grain yield

There was good agreement between simulated and observed forage and grain yield. The model simulated forage yield well ($r^2 = 0.70$, RMSE = 1.5 Mg ha^{-1} y^{-1}, rRMSE = 18%), capturing both high and low yields, with a mild tendency to overestimate forage production in the 6 to 12 Mg ha^{-1} y^{-1} range (Fig. 2). Soybean and sorghum grain yields were satisfactorily simulated (respectively $r^2 = 0.47$, RMSE = 0.6 Mg ha^{-1}, and $r^2 = 0.47$, RMSE = 1.0 Mg ha^{-1}), although three simulated soybean yields were much higher than those observed. These low yields correspond to the first two seasons of soybean in the long-term experiment, when insect damage reduced yield. It is noteworthy that each pairing of

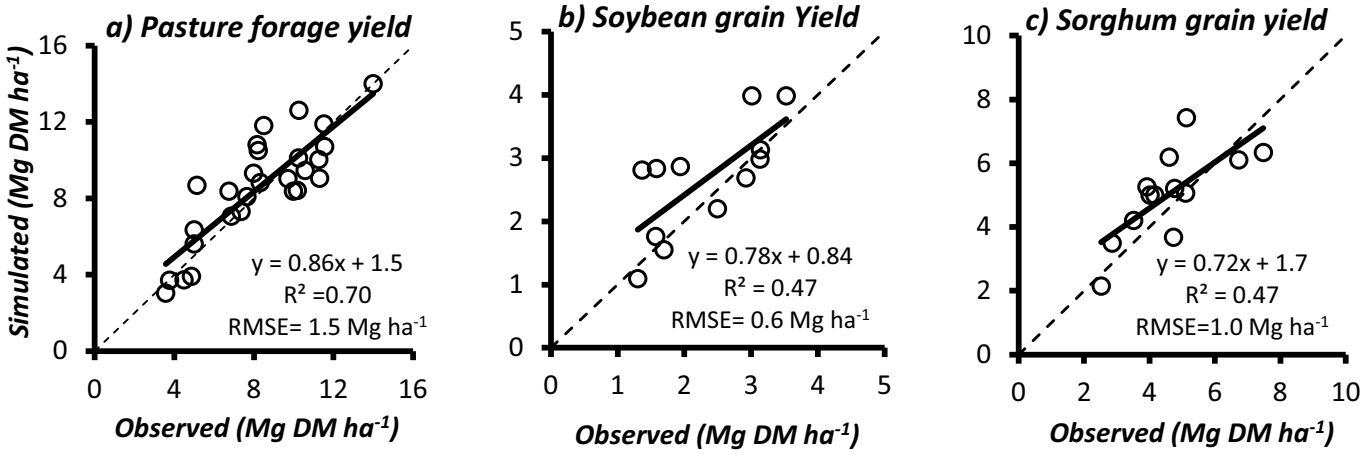


Fig. 2. Comparison of simulated against observed pasture yield from long term experiments at Palo a Pique experimental unit (INIA Treinta y Tres, Uruguay), showing the 1:1 line and fitted regression.

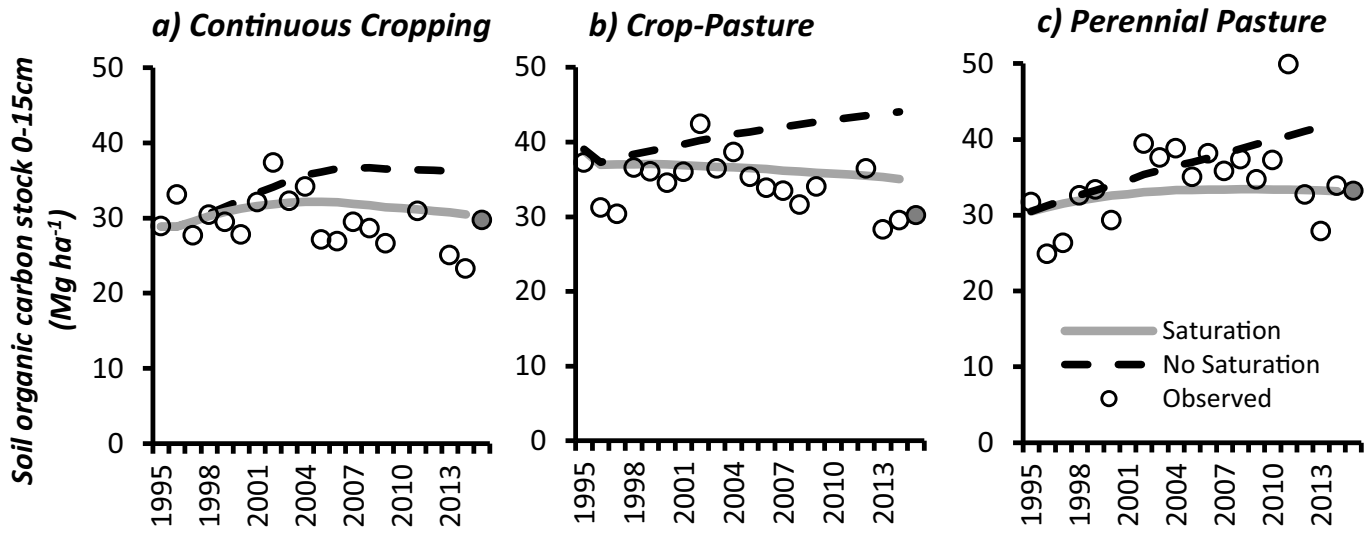


Fig. 3. Observed and modeled soil organic carbon mass in the top 15-cm of soil for three rotational systems in Treinta y Tres, Uruguay. Saturation and no saturation refer to the inclusion or not of the soil carbon saturation algorithm in the Cycles model. The closed circle in 2014 comes from an independent soil sampling and averages 40 profiles.

observed and simulated yield was obtained from the actual rotation sequence and reflects both the weather conditions during the corresponding growing season and the cumulative effects due to the corresponding rotation, as the model was not re-initialized every year. Considering the simulated forage and grain yield as proxies of total biomass production, the results indicate that the model represented well the C inputs to the soil. The multispecies module allowed Cycles to simulate mixed pastures yield with $r^2 = 0.7$ and RMSE 1.5 Mg ha^{-1}, while O'Leary et al. (2016) reported an $r^2 = 0.48$ and RMSE 2.8 Mg ha^{-1} y^{-1} for pastures growth simulated over eight years using the APSIM-Agpasture module in Hamilton, Victoria (Australia).

3.2. Soil C stocks

The C_s mass balance during the 1995–2014 period showed two different phases distinguishable for all systems (Fig. 3), and a time by system interaction (Table 4). In the first phase (1995 to 2004), C_s increased in all systems. In the second phase (2005 to 2014), C_s decreased in the systems with annual crops in the rotation (Table 5). The C_s mass accrual rates in the first period were 0.6 and 0.7 for CC and CP, and 1.3 Mg ha^{-1} y^{-1} for PP. In the second period, CC and CP lost C_s at a similar rate (−0.7 and − 0.6 Mg ha^{-1} y^{-1}, respectively), while PP kept C_s mass stable. Over the 20-year period, PP was the only system that accrued C_s (0.4 Mg ha^{-1} y^{-1}); CC and CP lost C_s at overall rates of −0.3 and − 0.2 Mg ha^{-1} y^{-1} (Table 5).

Although the measured data show high variability among samples, the simulations tracked reasonably well the general temporal trend of C_s mass for each rotational system (Fig. 3). The CC lost C_s mass in the second half of the period considered, while that of PP was stable. In these cases, the C_s from the last sampling (closed circle in Fig. 3) agreed well with the simulation result. For the CP rotation, both measured and modeled data show a mild C_s mass loss. Owing to a higher *h*, the no-saturation model accumulated C in the top 15 cm. When compared to measured data in the top 15-cm of soil, Cycles simulations with the saturation algorithm showed a better fit than the no-saturation algorithm. Considering all rotational systems, the saturation algorithm had a better fit than the non-saturation algorithm, with a RMSE of 4.7 and 6.8 Mg ha^{-1}, and rRMSE of 14% and 21%, respectively (Table 6). The RMSE obtained with the saturation algorithm in Cycles compares well with the RMSE of the statistical model ($17.5^{1/2} = 4.2$ Mg ha^{-1}, Table 4).

When considering the entire soil profile, both the model with and without saturation predicted losses of C in the 20-year period, but these C losses were much larger in the model without saturation (Table 7). In addition, the losses were distributed differently with depth. Comparing initial C_s contents to that in the 0–60 cm soil layer from 2015 sampling at 15-cm depth interval (Fig. 4), C_s shows increasing stratification only in PP, with greater topsoil C_s compared to CP and CC (19.6, 14.9 and 12.1 g C soil kg^{-1} respectively, $P < .10$). In contrast, the model with the non-saturation algorithm predicted an increase in C_s in the top soil for all systems, and sharp losses in the subsoil. Consequently, final C_s distribution in the soil profile over all systems was better for the model including the saturation algorithm than for the no-saturation algorithm, with an RMSE of 1.5 vs 2.1 g C soil kg^{-1}, respectively.

Table 4
Analysis of variance of the top 15 cm of soil carbon observed stocks for the period 1995–2014.

Effect	Degrees of freedom	F value	Pr(> F)
System	2	10.2	< 0.01
Year	1	5.4	< 0.03
Year quadratic term	1	29.9	< 0.01
System-year interaction	2	6.4	< 0.01
System-year interaction quadratic term	2	0.3	< 0.75
Error term (17.5 Mg^2 ha^{-2})	120		

3.3. Nitrogen balance

All systems lost N throughout the simulation period, in the order of 10 to 30 kg ha^{-1} y^{-1} of N for CC and CP and 3 kg ha^{-1} y^{-1} of N for PP (Table 8). Due to greater decomposition rate in deep soil layers, the losses of N in the simulation with the no-saturation algorithm were larger, in the order of 40 kg ha^{-1} y^{-1} of N for CC and CP and 10 kg ha^{-1} y^{-1} of N for PP. In CC, synthetic fertilizer was the main N input, while in the two systems with pastures it was BNF. The dominant losses were grain and forage harvest in CC, and N gaseous losses in the systems with pastures. Internal fluxes of N mineralization and immobilization were lower in CC, and increased with the pasture component in the rotation (Table 8).

Table 5
Rate of change of soil carbon mass (dC_s/dt) in the top 15-cm of soil for the three long-term rotation systems between 1995 and 2014. Saturation and no saturation refer to the inclusion or not of the soil carbon saturation algorithm in the Cycles model.

System	Observed dC_s/dt	p > F	Modeled dC_s/dt Saturation	Modeled dC_s/dt No saturation
	Mg ha^{-1} y^{-1}		Mg ha^{-1} y^{-1}	
Period 1995–2004				
Continuous Cropping	0.57	< 0.09	0.37	0.76
Crop-Pasture	0.65	< 0.09	−0.03	0.42
Perennial Pasture	1.34	< 0.01	0.25	0.59
Period 2004–2014				
Continuous Cropping	−0.66	< 0.06	−0.17	0.01
Crop-Pasture	−0.62	< 0.05	−0.15	0.30
Perennial Pasture	−0.37	< 0.50	−0.01	0.58

Table 6
Comparison of RMSE for observed and modeled soil organic carbon in the top 15 cm of soil for three rotational systems. Saturation and no saturation refers to the inclusion or not of the soil carbon saturation algorithm in the Cycles model.

	RMSE		rRMSE	
	Model		Model	
System	Saturation	No saturation	Saturation	No saturation
	Mg ha^{-1}		%	
Continuous Cropping	3.8	6.8	13	21
Crop-Pasture	5.0	8.0	14	23
Perennial Pasture	5.3	5.5	15	19
Mean	**4.7**	**6.8**	**14**	**21**

Table 7
Soil organic carbon (C_s) mass for the soil profile (0–1.9 m) at the beginning and percentage change at the end of the 20-year simulation for three rotational systems in Treinta y Tres, Uruguay. Saturation and no saturation refers to the inclusion or not of the soil carbon saturation algorithm in the Cycles model.

	Initial profile C_s mass	Twenty-years change in C_s	
		Model	
System		Saturation	No saturation
	Mg ha^{-1}	% of initial	
Continuous Cropping	96	−3.7	−9.2
Crop-Pasture	109	−6.5	−8.7
Perennial Pasture	100	−0.7	−3.0

4. Discussion

In the two decades and for the top 15-cm of soil, CC systems lost C_s at a rate of −0.3 Mg ha^{-1} y^{-1}, while PP systems accrued C_s at a rate of 0.4 Mg ha^{-1} y^{-1} (Fig. 3). Somewhat greater losses have been reported in tilled systems on Typic Argiudolls of Uruguay (García-Préchac et al., 2004), where a CC system lost C_s at a rate of −0.5 Mg ha^{-1} y^{-1} of C after 26 years, while an integrated crop-pasture system was essentially at equilibrium. The C_s accrual rate observed in the PP system was comparable with that of 0.6 Mg ha^{-1} y^{-1} reported by Franzluebbers and Stuedemann (2010) in 12-year pastures in the southeastern USA (Georgia), but lower than the accrual rate of 2.2–3.6 Mg ha^{-1} y^{-1} of C reported for a 18-year re-established grassland in Australia (Bell et al., 2012). The difference in the rate of change of C_s between CC and PP was 0.7 Mg ha^{-1} y^{-1} of C, very similar to the 0.84 ± 0.11 Mg ha^{-1} y^{-1} of C reported by Franzluebbers (2010) in a compilation of data for the southeastern USA.

The changes in C_s mass through time were well simulated with Cycles, with a RMSE of 4.7 Mg ha^{-1} of C and a rRMSE of 14% (Table 6). The RMSE compares well with the RMSE of 4.2 Mg ha^{-1} of C obtained from measurements (Table 4). Sharp year-to-year variations in measured C_s mass are most likely due to year to year changes in soil sampling depth and soil bulk density, indicating that it can be difficult and costly to track short-term variation in C_s mass via soil sampling. Cycles performance at simulating the changes in C_s through time compares well with that obtained with other agroecosystem models used in grassland or pasture soils (Table 9). The 14% rRMSE that we obtained is similar to the 16% rRMSE obtained by Cerri et al. (2004) in Brazil, but greater than the 5% obtained by O'Leary et al. (2016) for a thicker and therefore less variable soil layer (0–30 cm).

Potential losses of soil and C_s mass via erosion were not accounted for in the simulations. In the second half of the period considered, erosion could have been higher in CC than in other systems due to the higher frequency of soybean. The RUSLE-based (Wischmeier and Smith, 1978; Renard et al., 1997) software EROSION 5.9 (García-Préchac et al., 2005) calibrated for Uruguay with data from runoff plots located at the UEPP (Terra and García-Préchac, 2001), returns average soil losses of 13 Mg ha^{-1} y^{-1} for a soybean-cover crop sequence. Multiplying this soil loss by the initial topsoil C_s concentration on CC returns C erosion losses of 0.43 Mg ha^{-1} y^{-1}, matching observed C_s losses in CC. Although the RUSLE simulation has a higher frequency of soybean than our Cycles simulations, erosion may explain why the simulated C_s mass losses by Cycles seem to be lower than those observed (Fig. 3). While the variability in the measured C_s exceeds the magnitude of the calculated erosion, modeled changes in C_s storage rate need to account for erosion when soybean or bare soils occupy are a large portion of the rotation.

The control of both the humification rate and turnover rate through saturation in Cycles rendered a realistic representation of C_s gain and loss throughout the soil profile, preserving C_s in deep soil layers. When the model used the non-saturation algorithm, the estimated C_s concentration with depth drifted away from that observed (Fig. 4). There are two main controls in the saturation algorithm that explain this result. First, the reduction in h (the effective humification rate) in the topsoil due to higher saturation ratio (observed C_s/C_x range 0.67–0.87) prevented further C_s accretion. Second, the reduction in k (the Cs decay rate) in the subsoil due to lower saturation (C_s/C_x < 0.4) slowed down C_s decomposition and preserved the quasi-exponential decrease in C_s concentration with depth typically observed in soils (Fig. 4). Other agroecosystem models force this decrease in the C_s concentration with depth through forcing functions or by defining a layer-specific k (Oleson et al., 2013). In contrast, the decrease in C_s concentration with depth in Cycles is an emergent property of the saturation algorithm. One could argue that reducing h in the non-saturating model would have prevented the simulated C_s accumulation in the topsoil. While this is true, this would have accelerated the loss of C in the subsoil even more. Thus, it seems that the consideration of C saturation as a variable controlling not only h (Hassink and Whitmore, 1997; Stewart et al.,

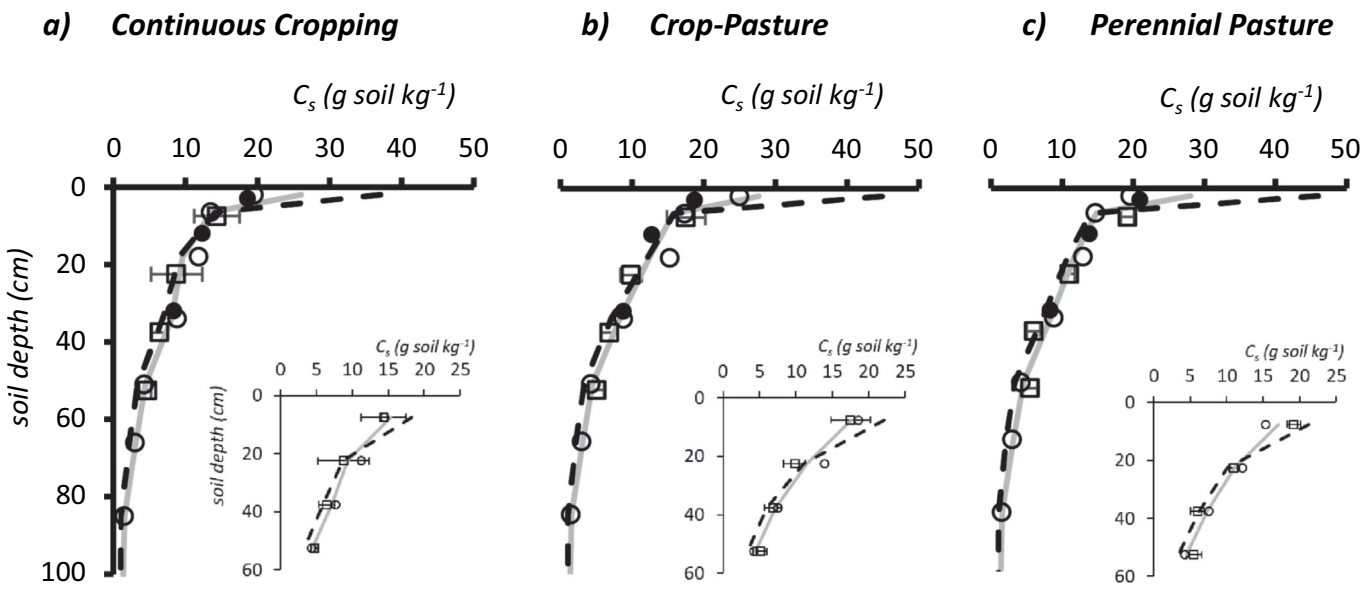


Fig. 4. Soil organic carbon distribution in the soil profile for the three rotation systems evaluated. Open circles represent C_s content in the soil profile survey in 1996. The closed circles represent C_s from samples taken on soil diagnostic horizons at the end of the simulation period, in 2014. Closed squares represent observed C_s on samples taken every 15 cm depth increments by Macedo and Terra in 2015. Grey solid lines represent C_s at the end of the 20 y simulation with Cycles model including saturation. Black broken lines represent C_s at the end of the 20 y simulation with no saturation algorithm. Inset figures show data at 15 cm increments up to 60 cm of soil depth.

Table 8
Simulated nitrogen balance. All components expressed in kg ha^{-1} y^{-1} of N. Saturation and no saturation refers to the inclusion or not of the soil carbon saturation algorithm in the Cycles model.

	Saturation model			No Saturation model		
System	CC	CP	PP	CC	CP	PP
	kg ha^{-1} y^{-1}					
N inputs						
Mean N fertilizer inputs	78	28	0	78	28	0
Biological Nitrogen Fixation (BNF)	34	56	72	31	52	64
Total inputs	112	85	72	109	81	64
N fluxes						
N mineralization	258	320	309	332	355	355
N immobilization	60	61	75	84	84	101
Net N mineralization	198	259	234	248	271	254
N outputs						
N leaching	6	5	0.1	17	10	1
N gaseous losses	41	59	75	47	57	72
N in harvest	77	50	0	82	50	0
Total outputs	124	114	75	147	117	73
Balance	−12	−29	−3	−38	−36	−9

CC: continuous cropping rotation of two years; CP: six-year crop-pasture rotation that alternates four years of grass-legume pasture with two years of annual crops; PP: perennial pasture of introduced legumes and grasses.

2008), but also *k*, is a necessary model property and a useful innovation in Cycles to accurately predict soil profile dynamics without needless parameterization.

Inputs of N in the three systems ranged from a maximum modeled estimate of 112 kg ha^{-1} y^{-1} of N with 70% as synthetic fertilizer, to a minimum of 72 kg ha^{-1} y^{-1} on PP relying only on legumes BNF. The N recovery efficiency calculated as N harvested divided by N inputs was 0.7 and 0.53 for CC and CP. These efficiencies compare well with a mean of 0.35 (range 0.20–0.49) in the European Union (Godinot et al., 2016) and 0.5 in US (Conant et al., 2013), and with a mean of 0.37 for fertilizer N inputs in Midwest maize systems (Cassman et al., 2002). Considering only N fertilizer inputs, the ratio of N recovered in harvest to synthetic N fertilization was 0.99 and 1.72 for CC and CP. This is important because N losses to groundwater and surface waters pollution depends on N surpluses, i.e. N additions exceeding N removals with harvest (McLellan et al., 2018), and N surpluses are inversely related to the fraction of N inputs from BNF (Blesh and Drinkwater, 2013). While livestock systems can become a source of N pollution for groundwater (Sakadevan and Nguyen, 2017), N leaching is mostly favored in grass-legume swards that receive high N fertilization (> 200 kg N ha^{-1} y^{-1}) in addition to BNF (Stout et al., 2000; Ledgard, 2001). In contrast, average N losses are generally low in grazed pastures relying on BNF (Ledgard, 2001). Accordingly, the modeled N leaching losses were low in the three systems (Table 8).

The simulated gaseous losses of N as N_2O were low (range 1.4–3.2 kg ha^{-1} y^{-1}), while the total gaseous losses of N were large. Our simulations show relatively large ammonia volatilization losses of 35 and 74 kg ha^{-1} y^{-1} of N in CC and PP. These losses increased with the proportion of pastures in the rotation and derived from animal urine patches. These losses can be high on a patch basis (170 kg ha^{-1} y^{-1} of N, Ledgard, 2001), but are restricted to the urine patch; averages over a paddock are much lower. For temperate South America, Piñeiro et al. (2010) reported ammonia losses of 17 to 25 kg ha^{-1} y^{-1} of N, somewhat higher than reports of < 17 kg ha^{-1} y^{-1} on grazed grass-legume swards by Ledgard (2001) and Saarijärvi et al. (2006). Mulvaney et al. (2008) estimated annual ammonia volatilization of 20 kg ha^{-1} y^{-1} of N for dairy cows at a stocking density of 4 cattle ha^{-1} (range 5 to 48 kg ha^{-1} y^{-1} of N). In our simulations, higher ammonia volatilization losses were somewhat compensated for lower than expected denitrification losses (range 2 to 9 kg ha^{-1} y^{-1} of N). Nonetheless, while the simulated ammonia losses seem to be high, they did not limit the pasture productivity due to the legumes BNF in the simulated pastures. For the same reason, these losses may indirectly cause higher than expected simulated BNF. This topic requires further experimental and modeling work.

Both N mineralization and immobilization fluxes increased along with an increase in the pasture proportion in the rotation (Table 8). This result is consistent with the hypothesis that as C_s approaches saturation, the increase in organic matter also produces an increase in N turnover

Table 9
Reported average performance of different models with respect to the prediction of soil organic carbon for pasture systems as determined by the root mean square error (RMSE) and relative root mean square error (rRMSE), showing the location, temperature and precipitation at the site, and soil depth evaluated.

Model	Location	Mean Temp.	Annual Precipitation	Soil depth	RMSE	rRMSE	Reference
		– °C –	– mm y^{-1} –	– cm –	- Mg ha^{-1} -	– % –	
Cycles	Treinta y Tres, Uruguay	17	1300	0–15	4.7	14	This Publication
APSIM-AgPasture	Hamilton, Victoria, Australia	13	690	0–30	4.9	5	O'Leary et al. (2016)
RothC	Waga Waga, NSW Australia	16	650	0–30	3.2	8	Liu et al. (2011)
RothC – after inverse modeling	Waga Waga, NSW Australia	16	650	0–30	2.1	5	Liu et al. (2011)
Century	Rondônia, Brazil	26	2200	0–20	NR	16	Cerri et al. (2004)

References: Temp.: Temperature, NR: Not reported

associated to the progressive saturation of N sinks in the system (Aber et al., 1989; Lovett and Goodale, 2011). Aber et al. (1989) stated that as a soil approaches N saturation, an increase in N mineralization is observed before N losses increase. As a result from an increase in turnover, internal N fluxes may be large whithout N losses increasing proportionally. Therefore, there seems to be an optimal range for C_s saturation ratio that ensures N supply to the plant but limits N losses.

In a framework of soil organic matter cycling centered around C_s, the accrual and loss of N_s can be regulated in two ways: they can be tighthly coupled to the accrual and loss of C_s through a fixed soil organic matter C:N ratio, or they can be partially decoupled through changes in the pool C:N ratio (Baer and Blair, 2008; Lovett and Goodale, 2011). Although soil C:N ratios are relatively well constrained (Cleveland and Liptzin, 2007), small variations in this ratio can both buffer N_s losses and support large N mineralization from soil organic matter and residues. In Cycles, h decreases and k increases as C_s approaches the saturation ratio, a mathematical representation of the lesser ability of the soil physical matrix to protect organic substrates from microbial decomposition (eqs. 2–4, Fig. 1). Taken together, lower C retention and higher turnover rate can provide an optimum range that favors N mineralization, but that is self-restrained by immediate increases in h and reductions in k once saturation decreases. The close-to-saturation condition in the topsoil (C_s/C_x of 0.67 to 0.87) in the crop pasture systems seems to enable faster C and N turnover, gradually supplying N from an organic pool, thus minimizing leaching losses.

The simulation results agree with the expectation of a declining N supply when converting crop-pasture systems to continuous cropping in no-till systems of low inputs. The net N mineralization decreased by 61 kg N ha^{-1} y^{-1} in CC compared to CP when running the model with the saturation algorithm (Table 8). If this decline were to be compensated by N fertilizer, external N inputs in CC should be increased at least by 79%. This is consistent with a report by Ernst et al. (2016), who attributed part of the grain yield decline in wheat to a declining N supply when the annual cropping phase lengthens. It has been stated that integrated crop-livestock systems can recouple C and N cycling and play a critical role in sustainable intensification (Carvalho et al., 2010; Soussana and Lemaire, 2014). The basis for such proposition seems to be the C_s and N_s saturation dynamics, which enables both N supply and N retention at rates that satisfy crop demand but limit losses to the environment. The challenge for researchers and practitioners is to define in unambiguous terms the right saturation band. This is a topic that clearly merits further research.

The empirical relationships between both saturation and fresh C inputs retention rate (h) and saturation and C_s decomposition rate (k) used in Cycles need to be thoroughly tested. The current model parameterization relies on interpretation of limited data and modeling results (Hassink et al. 1997, Kemanian and Stöckle, 2010, Kemanian et al., 2011, White et al., 2014), and therefore it is of critical importance to design and execute experiments that can help generalize and refine our understanding of fresh residues C retention and C_s decomposition. Such experiments should provide quantitative proof of the changes in h and k as the saturation ratio changes, as well as quantitative evidence of fluctuations in the C:N ratio. For example, C_s saturated soil layers can allow further N_s accumulation by a progressive reduction in the C:N ratio of the soil organic matter. These experiments are needed to further validate and parameterize C_s and N_s cycling models that are based on C (and N) saturation theory.

5. Conclusions

Incorporating C_s saturation into an agroecosystems model allowed simulation of the long-term C_s changes in crop-pasture rotational systems, reproducing with fidelity the C_s distribution with depth. In addition, N turnover (mineralization-immobilization) accelerated when C_s saturation was higher, which occurred in systems with a higher proportion of pastures with perennial plants species in the rotation. Our research suggests that a possible reason for the high productivity and low environmental footprint of crop-pasture rotations might be that they operate within the C_s saturation band that provides N in synchrony with annual crops N demand, without facilitating N losses. In this way, crop-pasture rotational systems can be integral to the sustainable intensification of agricultural production in temperate and subtropical environments.

Acknowledgements

We appreciate the proficient assistance of INIA staff during the long-term experiment and soil sampling campaign. Funding for this work was provided by the Instituto Nacional de Investigación Agropecuaria (INIA-Uruguay) and the USDA-ARS Research Agreement Contract #58-1902-1-165 (Modeling of multispecies pasture growth and management).

Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2018.11.001.

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